

Adaptation Experiment 1: Univariate Control

1 Theoretical overview and goal

A forecasting adapter is useful only if it improves a well-defined, untouched backbone. For a look-back $x \in \mathbb{R}^L$, a pretrained forecaster produces $\hat{y}_0 = f(x) \in \mathbb{R}^H$. The univariate control evaluates f directly, without retrieved neighbors, gates, or trainable adaptation. It therefore separates backbone error from error introduced by the retrieval and adaptation pipeline.

The main metrics are

$$\text{MSE} = H^{-1} \|\hat{y}_0 - y\|_2^2, \quad \text{nMSE} = H^{-1} \left\| \frac{\hat{y}_0 - y}{\sigma(x) + \epsilon} \right\|_2^2.$$

Per-user aggregation and the worst 10% user mean reveal whether a good global mean hides systematic failures on a subset of series.

Experimental goal. Establish the no-adaptation reference, measure its temporal and user-wise heterogeneity, and provide the denominator against which every retrieval method must be compared.

2 Windows and chronological protocol

The integer s is always the query date and the last observed date. The look-back and target are

$$X_s = (s - L, s] = \{z_{s-L+1}, \dots, z_s\}, \quad Y_s = (s, s + H] = \{z_{s+1}, \dots, z_{s+H}\}.$$

They contain exactly L and H values; the look-back never contains the first forecast value. If a target interval begins at date b , its first query is $s = b - 1$. The look-back may cross that boundary, but the complete Y_s must lie inside the selected target interval. Consequently the held-out T_3 target dates do not shift when L changes.

Raw dates are first split chronologically as $(T_0, T_1 + T_2, T_3) = (0.30, 0.50, 0.20)$. Extraction writes pooled **adapt** ($T_1 + T_2$) and held-out **eval** (T_3) payloads. Each downstream model makes its own whole-query-date split of **adapt**; users sharing one query date can never be divided between T_1 and T_2 . The control evaluates T_3 . Each query contains all users at one date, but the backbone is called as a univariate model with one channel per sample.

Dataset preparation automatically reads the sibling `config.json`. Portable top-level options and adaptation-scoped options are combined before the chronological split; user exclusions are additive and the selected config path is logged. The same preparation contract is used by extraction, the univariate control, and all downstream adapters.

The test profile uses Electricity 168–24 with sparse evaluation. The small profile reproduces the three-dataset subset; full widens the Chronos control to six datasets and adds the Cross-RAG setting. Ultra has the same direct Chronos control because this reference launcher has no second backbone.

Component	Current sweep value
Datasets	Traffic, Electricity, Solar
Look-back–horizon	168–24, 504–24, 504–168, 504–504
Backbone	Chronos
Input normalization	Instance normalization
Evaluation stride	128 dates
Seed	1

Implementation: `src/experiments/experiment_univariate.py`

Launcher: `univariate.slurm`

It writes window-level losses, a JSON summary, a tensor payload, error distributions, horizon-error curves, user scatter plots, and prediction examples. The primary comparison is the same T_3 MSE/nMSE later used by adapted methods.

3 Interpretation

This experiment is descriptive rather than trainable. Any adapted method must beat the control on the same queries and aggregation rule. Improvements on T_1 or T_2 alone are insufficient evidence because those periods are used to fit adapters or gates.